

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Integrate each of the following.

a) $\int_{-\pi/2}^{\pi/2} \cos^7(x) dx$

$$\int_{-\pi/2}^{\pi/2} \cos^7(x) dx = \int_{-\pi/2}^{\pi/2} \cos^6(x) \cos(x) dx = \int_{-\pi/2}^{\pi/2} [1 - \sin^2(x)]^3 \cos(x) dx$$

Let $u = \sin(x)$ and $du = \cos(x) dx$. The bounds become $u\left(-\frac{\pi}{2}\right) = -1$ and $u\left(\frac{\pi}{2}\right) = 1$. Therefore, we have,

$$\begin{aligned} \int_{-\pi/2}^{\pi/2} \cos^7(x) dx &= \int_{-1}^1 [1 - u^2]^3 du = \int_{-1}^1 [1 - 3u^2 + 3u^4 - u^6] du = \left[u - u^3 + \frac{3}{5}u^5 - \frac{1}{7}u^7 \right]_{-1}^1 \\ &= \left[1 - 1 + \frac{3}{5} - \frac{1}{7} \right] - \left[-1 + 1 - \frac{3}{5} + \frac{1}{7} \right] = \frac{6}{5} - \frac{2}{7} = \frac{21}{35} - \frac{5}{35} \\ &= \frac{16}{35} \end{aligned}$$

b) $\int_{-\pi/2}^{\pi/2} \cos^7(x) \sin^3(x) dx$

$$\int_{-\pi/2}^{\pi/2} \cos^7(x) \sin^3(x) dx = \int_{-\pi/2}^{\pi/2} \cos^6(x) \sin^2(x) \sin(x) dx = \int_{-\pi/2}^{\pi/2} \cos^6(x) [1 - \cos^2(x)] \sin(x) dx$$

Let $u = \cos(x)$ and $du = -\sin(x) dx \rightarrow -du = \sin(x) dx$. The bounds become $u\left(-\frac{\pi}{2}\right) = 0$ and

$u\left(\frac{\pi}{2}\right) = 0$. Therefore, we have,

$$\int_{-\pi/2}^{\pi/2} \cos^7(x) \sin^3(x) dx = \int_{-\pi/2}^{\pi/2} \cos^6(x) [1 - \cos^2(x)] \sin(x) dx = \int_0^0 -u^6 [1 - u^2] du = 0$$

c) $\int_{-\pi/2}^{\pi/2} \cos^4(x) dx$

$$\begin{aligned} \int_{-\pi/2}^{\pi/2} \cos^4(x) dx &= \int_{-\pi/2}^{\pi/2} \cos^2(x) \cos^2(x) dx = \int_{-\pi/2}^{\pi/2} \frac{1}{2} [1 + \cos(2x)] \frac{1}{2} [1 + \cos(2x)] dx = \int_{-\pi/2}^{\pi/2} \frac{1}{4} [1 + \cos(2x)]^2 dx \\ &= \int_{-\pi/2}^{\pi/2} \frac{1}{4} [1 + 2\cos(2x) + \cos^2(2x)] dx = \int_{-\pi/2}^{\pi/2} \frac{1}{4} \left[1 + 2\cos(2x) + \frac{1}{2} [1 + \cos(4x)] \right] dx \\ &= \int_{-\pi/2}^{\pi/2} \frac{1}{4} \left[\frac{3}{2} + 2\cos(2x) + \frac{1}{2} \cos(4x) \right] dx = \frac{1}{4} \left[\frac{3}{2} x + \sin(2x) + \frac{1}{8} \sin(4x) \right] \Big|_{-\pi/2}^{\pi/2} \\ &= \frac{1}{4} \left[\frac{3}{2} \left(\frac{\pi}{2} \right) + \sin(\pi) + \frac{1}{8} \sin(2\pi) \right] - \frac{1}{4} \left[\frac{3}{2} \left(-\frac{\pi}{2} \right) + \sin(-\pi) + \frac{1}{8} \sin(-2\pi) \right] = \frac{3\pi}{16} + \frac{3\pi}{16} \\ &= \frac{3\pi}{8} \end{aligned}$$

d) $\int_0^{\pi/6} \sqrt{1 + \cos(2x)} dx$

We know that $\cos^2(x) = \frac{1}{2} [1 + \cos(2x)] \rightarrow 2\cos^2(x) = 1 + \cos(2x) \rightarrow \sqrt{2} \cos(x) = \pm \sqrt{1 + \cos(2x)}$.

Since we are integrating over the interval $\left[0, \frac{\pi}{6}\right]$ we see that $\sqrt{2} \cos(x)$ must be positive and therefore

$$\sqrt{2} \cos(x) = \sqrt{1 + \cos(2x)}.$$

This implies that,

$$\int_0^{\pi/6} \sqrt{1 + \cos(2x)} dx = \int_0^{\pi/6} \sqrt{2} \cos(x) dx = \sqrt{2} \sin(x) \Big|_0^{\pi/6} = \sqrt{2} \sin\left(\frac{\pi}{6}\right) - \sqrt{2} \sin(0) = \sqrt{2} \left(\frac{1}{2}\right) - \sqrt{2}(0) = \frac{\sqrt{2}}{2}$$

2. Evaluate each of the following integrals.

a) $\int \tan(x) dx$. (No Calc 2 techniques are needed here)

Frist, we rewrite the integral $\int \tan(x) dx = \int \frac{\sin(x)}{\cos(x)} dx$.

Let $u = \cos(x)$ then $du = -\sin(x) dx \rightarrow -du = \sin(x) dx$

Therefore,

$$\int \tan(x) dx = \int \frac{\sin(x)}{\cos(x)} dx = \int -\frac{1}{u} du = -\ln|u| + C = -\ln|\cos(x)| + C$$

b) $\int \tan^2(x) dx$

$$\int \tan^2(x) dx = \int [\sec^2(x) - 1] dx = \tan(x) - x + C$$

c) $\int \tan^3(x) dx$ (No Calc 2 techniques are needed here)

$$\begin{aligned} \int \tan^3(x) dx &= \int [1 - \sec^2(x)] \tan(x) dx = \int [\tan(x) - \tan(x) \sec^2(x)] dx \\ &= \int \tan(x) dx - \int \tan(x) \sec^2(x) dx \end{aligned}$$

For the second integral we let $u = \tan(x)$ and then $du = \sec^2(x) dx$. Therefore, we have,

$$\int \tan(x) dx - \int \tan(x) \sec^2(x) dx = \int \tan(x) dx - \int u du = -\ln|\cos(x)| - \frac{1}{2}u^2 + C = -\ln|\cos(x)| - \frac{1}{2}\tan^2(x) + C$$

d) $\int \tan^3(x) \sec^3(x) dx$

First, we write

$$\int \tan^3(x) \sec^3(x) dx = \int \tan^2(x) \sec^2(x) \sec(x) \tan(x) dx = \int [\sec^2(x) - 1] \sec^2(x) \sec(x) \tan(x) dx$$

Let $u = \sec(x)$ and then $du = \sec(x) \tan(x) dx$. Therefore,

$$\begin{aligned} \int [\sec^2(x) - 1] \sec^2(x) \sec(x) \tan(x) dx &= \int [u^2 - 1] u^2 du = \int [u^4 - u^2] du \\ &= \frac{1}{5}u^5 - \frac{1}{3}u^3 + C \\ &= \frac{1}{5}\sec^5(x) - \frac{1}{3}\sec^3(x) + C \end{aligned}$$

3. Consider the integral $\int \frac{10x}{\sqrt{3+x^2}} dx$.

a) Evaluate this integral.

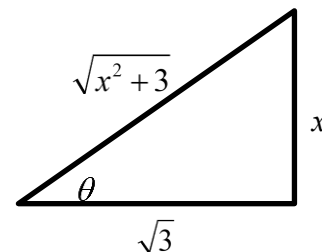
Let $x = \sqrt{3} \tan(\theta)$ where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, then $dx = \sqrt{3} \sec^2(\theta) d\theta$. Therefore,

$$\begin{aligned} \int \frac{10x}{\sqrt{3+x^2}} dx &= \int \frac{10\sqrt{3} \tan(\theta)}{\sqrt{3+(\sqrt{3} \tan(\theta))^2}} \sqrt{3} \sec^2(\theta) d\theta = \int \frac{30 \sec^2(\theta) \tan(\theta)}{\sqrt{3} \sqrt{\sec^2(\theta)}} d\theta \\ &= \int \frac{30 \sec^2(\theta) \tan(\theta)}{\sqrt{3} \sec(\theta)} d\theta = 10\sqrt{3} \int \sec(\theta) \tan(\theta) d\theta = 10\sqrt{3} \sec(\theta) + C \end{aligned}$$

We must now write our solution in terms of x . Since

$x = \sqrt{3} \tan(\theta) \rightarrow \tan(\theta) = \frac{x}{\sqrt{3}}$ and we can construct the following reference

triangle for θ (Note: this triangle is the same in the first and fourth quadrants) is:



From the reference triangle we see that $\sec(\theta) = \frac{\sqrt{x^2+3}}{\sqrt{3}}$ and therefore

$$10\sqrt{3} \sec(\theta) + C = 10\sqrt{3} \left(\frac{\sqrt{x^2+3}}{\sqrt{3}} \right) + C = 10\sqrt{x^2+3} + C$$

b) Would it have been okay to use the angle restriction of $\frac{3\pi}{2} < \theta < \frac{5\pi}{2}$? What, if anything, would this change about your solutions from part (a)?

No, this doesn't change anything about the solution here. The secant function is still always positive on this interval and so our radical can still be simplified in the same way.

c) Suppose that our antiderivative contained θ . Would it be possible to replace theta with $\tan^{-1}\left(\frac{x}{\sqrt{3}}\right)$?

YES! The inverse tangent function can only return angles that exist between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ and since we have θ restricted to be between these values then it is possible for $\theta = \tan^{-1}\left(\frac{x}{\sqrt{3}}\right)$

d) Suppose that our antiderivative contained θ . Would it be possible to replace theta with $\tan^{-1}\left(\frac{x}{\sqrt{3}}\right)$ if

we were using the angle restriction $\frac{3\pi}{2} < \theta < \frac{5\pi}{2}$?

NO! The inverse tangent function can only return angles that exist between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$ and since we have θ restricted to be between $\frac{3\pi}{2}$ and $\frac{5\pi}{2}$ then it is possible for $\theta = \tan^{-1}\left(\frac{x}{\sqrt{3}}\right)$.

4. Evaluate the integral $\int_1^3 \frac{10x}{\sqrt{3+x^2}} dx$.

Let $x = \sqrt{3} \tan(\theta)$ where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, then $dx = \sqrt{3} \sec^2(\theta) d\theta$.

Our bounds then become $1 = \sqrt{3} \tan(\theta) \rightarrow \tan(\theta) = \frac{1}{\sqrt{3}} \rightarrow \theta = \frac{\pi}{6}$ and

$$3 = \sqrt{3} \tan(\theta) \rightarrow \tan(\theta) = \frac{3}{\sqrt{3}} = \sqrt{3} \rightarrow \theta = \frac{\pi}{3}$$

Therefore,

$$\begin{aligned} \int_1^3 \frac{10x}{\sqrt{3+x^2}} dx &= \int_{\pi/6}^{\pi/3} \frac{10\sqrt{3} \tan(\theta)}{\sqrt{3+(\sqrt{3} \tan(\theta))^2}} \sqrt{3} \sec^2(\theta) d\theta = \int_{\pi/6}^{\pi/3} \frac{30 \sec^2(\theta) \tan(\theta)}{\sqrt{3} \sqrt{\sec^2(\theta)}} d\theta \\ &= \int_{\pi/6}^{\pi/3} \frac{30 \sec^2(\theta) \tan(\theta)}{\sqrt{3} \sec(\theta)} d\theta = 10\sqrt{3} \int_{\pi/6}^{\pi/3} \sec(\theta) \tan(\theta) d\theta = 10\sqrt{3} \sec(\theta) \Big|_{\pi/6}^{\pi/3} \\ &= 10\sqrt{3} \sec\left(\frac{\pi}{3}\right) - 10\sqrt{3} \sec\left(\frac{\pi}{6}\right) = 10\sqrt{3} \left[2 - \frac{2}{\sqrt{3}}\right] = 20\sqrt{3} - 20 \end{aligned}$$

5. Evaluate the definite integral $\int_0^2 x\sqrt{1-(x-1)^2} dx$.

Let $x-1 = \sin(\theta)$ where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Then, $dx = \cos(\theta) d\theta$ and the bounds of our integral become

$0-1 = \sin(\theta) \rightarrow \theta = -\frac{\pi}{2}$ & $2-1 = \sin(\theta) \rightarrow \theta = \frac{\pi}{2}$. Therefore,

$$\begin{aligned} \int_0^2 x\sqrt{1-(x-1)^2} dx &= \int_{-\pi/2}^{\pi/2} [\sin(\theta)+1]\sqrt{1-(\sin(\theta))^2} \cos(\theta) d\theta \\ &= \int_{-\pi/2}^{\pi/2} [\sin(\theta)+1]\sqrt{\cos^2(\theta)} \cos(\theta) d\theta \\ &= \int_{-\pi/2}^{\pi/2} [\sin(\theta)+1]\cos^2(\theta) d\theta \\ &= \int_{-\pi/2}^{\pi/2} [\sin(\theta)\cos^2(\theta) + \cos^2(\theta)] d\theta \\ &= \int_{-\pi/2}^{\pi/2} \sin(\theta)\cos^2(\theta) d\theta + \int_{-\pi/2}^{\pi/2} \cos^2(\theta) d\theta \end{aligned}$$

For the first integral we let $u = \cos(\theta)$ and then $du = -\sin(\theta) d\theta \rightarrow -du = \sin(\theta) d\theta$ and the bounds become $u\left(-\frac{\pi}{2}\right) = \cos\left(-\frac{\pi}{2}\right) = 0$ and $u\left(\frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2}\right) = 0$. For the 2nd integral we use a trig identity.

$$\begin{aligned} \int_{-\pi/2}^{\pi/2} \sin(\theta)\cos^2(\theta) d\theta + \int_{-\pi/2}^{\pi/2} \cos^2(\theta) d\theta &= \int_0^0 -u^2 du + \int_{-\pi/2}^{\pi/2} \frac{1}{2}[1+\cos(2\theta)] d\theta \\ &= 0 + \frac{1}{2} \left[\theta + \frac{1}{2} \sin(2\theta) \right]_{-\pi/2}^{\pi/2} \\ &= \frac{1}{2} \left[\frac{\pi}{2} + \frac{1}{2} \sin(\pi) \right] - \frac{1}{2} \left[-\frac{\pi}{2} + \frac{1}{2} \sin(-\pi) \right] \\ &= \frac{1}{2} \left[\frac{\pi}{2} + 0 \right] - \frac{1}{2} \left[-\frac{\pi}{2} + 0 \right] \\ &= \pi \end{aligned}$$

6. Evaluate the integral $\int_2^4 \sqrt{(x-2)(4-x)} dx$.

$$\int_2^4 \sqrt{(x-2)(4-x)} dx = \int_2^4 \sqrt{-x^2 + 6x - 8} dx = \int_2^4 \sqrt{-(x^2 - 6x + 9) + 9 - 8} dx = \int_2^4 \sqrt{-(x-3)^2 + 1} dx$$

Let $x-3 = \sin(\theta)$ where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Then, $dx = \cos(\theta) d\theta$ and the bounds of our integral become

$$2-3 = \sin(\theta) \rightarrow \theta = -\frac{\pi}{2} \quad \& \quad 4-3 = \sin(\theta) \rightarrow \theta = \frac{\pi}{2}. \text{ Therefore,}$$

$$\begin{aligned} \int_2^4 \sqrt{-(x-3)^2 + 1} dx &= \int_{-\pi/2}^{\pi/2} \sqrt{-(\sin(\theta))^2 + 1} \cos(\theta) d\theta = \int_{-\pi/2}^{\pi/2} \sqrt{\cos^2(\theta)} \cos(\theta) d\theta \\ &= \int_{-\pi/2}^{\pi/2} \cos^2(\theta) d\theta = \int_{-\pi/2}^{\pi/2} \frac{1}{2} [1 + \cos(2\theta)] d\theta = \frac{1}{2} \left[\theta + \frac{1}{2} \sin(2\theta) \right] \Big|_{-\pi/2}^{\pi/2} \\ &= \frac{1}{2} \left[\frac{\pi}{2} + \frac{1}{2} \sin(\pi) \right] - \frac{1}{2} \left[-\frac{\pi}{2} + \frac{1}{2} \sin(-\pi) \right] \\ &= \frac{1}{2} \left[\frac{\pi}{2} + 0 \right] - \frac{1}{2} \left[-\frac{\pi}{2} + 0 \right] \\ &= \pi \end{aligned}$$